Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 18)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
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6	
7	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Which of the following best describes a **representative** sample? Tick one box.

[1 mark]

A A sample taken from the most convenient location ☐

B A sub-sample that accurately reflects the complete data set ☐

C The largest sample that can be collected ☐

D A sample that supports the hypothesis ☐

0 1 . 2 Describe how random sampling sites would be selected in an irregularly shaped meadow.

[2 marks]

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0 1 . 3 Explain why deliberately choosing sampling sites introduces bias.

[2 marks]

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0 2 . 1 For each of the following forms of direct exploitation, give a named species whose population has been reduced.

[5 marks]

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Food .....

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Fashion .....

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Entertainment / pets .....

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Furniture and ornaments .....

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Traditional medicines .....

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0 2 . 2 Explain why species are deliberately eradicated. Give **three** categories with examples.

[3 marks]

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0 2 . 3 Explain why legal protection alone rarely prevents the decline of a species exploited for traditional medicine.

[2 marks]

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0 3 . 1 Explain how overfishing of the giant triton mollusc damaged coral reefs in Australia.

[2 marks]

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0 3 . 2 Describe **four** management practices used in marine protected areas to protect coral reefs.

[4 marks]

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0 3 . 3 Explain why deep-water coral reefs recover much more slowly from trawling damage than tropical reefs.

[2 marks]

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0 3 . 4 Give **two** threats to deep-water reefs and **one** conservation measure.

[2 marks]

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0 4 Table 1 compares an indigenous mixed broadleaf woodland with a Sitka spruce plantation of the same area in Scotland.

Table 1

Feature	Mixed indigenous woodland	Sitka spruce plantation
Tree species	12	1
Age structure	All ages, dead wood present	All planted in the same year
Planting density	Natural, variable	Close, 2 m spacing
Timber yield / m^3 ha^{-1} yr^{-1}	4	16
Bird species	38	9
Ground flora species	60	4

0 4 . 1 Use Table 1 to explain why the plantation has a much higher timber yield but far lower biodiversity.

[6 marks]

0 4 . 2 Explain why commercial plantations are felled before the trees reach full size.

[2 marks]

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0 4 . 3 Explain why there is little demand today for coppiced or pollarded wood.

[2 marks]

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0 5 . 1 Explain why the ecosystem services provided by other species are often given a low conservation priority.

[2 marks]

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0 5 . 2 Explain how soil organisms provide services that humans depend on.

[3 marks]

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0 5 . 3 Explain why pollination by insects is more useful than wind pollination to plants growing in forests.

[2 marks]

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0 5 . 4 Explain how seed dispersal by animals increases the survival of a plant's offspring.

[3 marks]

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0 6 . 1 Describe coppicing and explain how it increases the biodiversity of a woodland.

[4 marks]

0 6 . 2 Explain how the reintroduction of grey wolves to Yellowstone National Park increased biodiversity.

[4 marks]

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0 6 . 3 Explain why culling or removing species may be necessary in habitat management.

[2 marks]

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0 7 Table 1 shows some events in the history of life on Earth.

Table 1

Event	Time before present / billion years
Formation of the Earth	4.6
First life (Archaea) on Earth	3.5
First photosynthetic organisms	2.7
Oxygen concentration in the atmosphere starts to rise	2.45
End of the period of oxygen build-up (Proterozoic)	0.54

0 7 . 1 Use Table 1 to calculate how many **million** years passed between the appearance of photosynthetic organisms and the start of the rise in atmospheric oxygen.

[1 mark]

Answer [ ] million years

0 7 . 2 Explain why atmospheric oxygen did not start to rise as soon as photosynthesis began.

[2 marks]

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0 7 . 3 Explain how the rise in atmospheric oxygen made life on the land surface possible.

[3 marks]

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0 7 . 4 Explain how early photosynthetic organisms helped to prevent a long-term rise in the Earth's temperature.

[2 marks]

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0 7 . 5 Give **two** other ways in which living organisms changed the conditions on Earth.

[2 marks]

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0 8 . 1 Explain how natural material cycles differ from linear human systems of material use.

[4 marks]

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0 8 . 2 Explain why the use of toxic metals in electronic products is a particular problem for sustainability.

[2 marks]

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0 8 . 3 Describe how the design of a product can move it from a linear to a cyclical system.

[4 marks]

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0 9 . 1 Give **two** habitats created by farming that support important wildlife communities.

[2 marks]

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0 9 . 2 Explain how drainage and nutrient enrichment of farmland reduce biodiversity.

[2 marks]

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0 9 . 3 Explain **one** problem that can be caused by introducing a biological control species.

[1 mark]

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1 0 . 1 State the Brundtland Commission definition of sustainable development.

[1 mark]

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1 0 . 2 Explain the difference between a negative feedback mechanism and a positive feedback mechanism, giving **one** example of each in the climate system.

[4 marks]

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1 0 . 3 Explain why a diverse ecosystem is more likely to resist change than an agroecosystem.

[1 mark]

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1 0 . 4 Evaluate the extent to which copying the principles of natural systems could make human activities more sustainable.

[9 marks]

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Extra space

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Write an essay on one of the following topics.

1 1 . 1 Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness.

[25 marks]

OR

1 1 . 2 Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS